

$$I = c_1 \int_{\epsilon N^{2/3}}^{\infty} \frac{1}{s} e^{-c_2 s^{2/3}} ds$$

Substitute $u = s^{3/2}$ s.t. $du = \frac{3}{2} s^{1/2} ds = \frac{3}{2} u^{1/3} ds$; Limits $\epsilon N^{2/3} \rightarrow \epsilon^{3/2} N$, $\infty \rightarrow \infty$

$$I = c_1 \int_{\epsilon^{3/2} N}^{\infty} \frac{1}{u^{2/3}} e^{-c_2 u} \cdot \frac{2}{3} \cdot \frac{1}{u^{1/3}} du$$

$$= \frac{2}{3} c_1 \int_{\epsilon^{3/2} N}^{\infty} \frac{e^{-c_2 u}}{u} du$$

Then via IBP:

$$u = \frac{1}{v} \quad v' = e^{-c_2 u}$$

$$u' = -\frac{1}{u^2} \quad v = -\frac{1}{c_2} e^{-c_2 u}$$

$$\therefore I = \frac{2}{3} c_1 \left(\underbrace{\left[-\frac{1}{c_2 u} e^{-c_2 u} \right]_{\epsilon^{3/2} N}^{\infty}}_{\phi_1} - \underbrace{\int_{\epsilon^{3/2} N}^{\infty} \frac{1}{c_2 u^2} e^{-c_2 u} du}_{\phi_2} \right)$$

$$\phi_1 = \left[-\frac{1}{c_2 u} e^{-c_2 u} \right]_{\epsilon^{3/2} N}^{\infty} = 0 + \frac{1}{c_2 \epsilon^{3/2} N} e^{-c_2 \epsilon^{3/2} N}$$

ϕ_2 can be evaluated similarly to before, and will yield

$$p = -\frac{1}{u^2} \quad q' = e^{-c_2 u}$$

$$p' = \frac{2}{u^3} \quad q = -\frac{1}{c_2} e^{-c_2 u}$$

$$\therefore \phi_2 = \left[\left[-\frac{1}{c_2 u^2} e^{-c_2 u} \right]_{\epsilon^{3/2} N}^{\infty} + \int_{\epsilon^{3/2} N}^{\infty} \frac{2}{c_2 u^3} e^{-c_2 u} du \right]$$

Consider the u^2 term in the above (and then subsequent $u^3, u^4, \dots$) that we would find, as $N \rightarrow \infty$, these terms will be dominated by ϕ_1 . Hence, we can say that in the large N limit

$$I = \frac{2}{3} c_1 \cdot \frac{1}{c_2 \epsilon^{3/2} N} e^{-c_2 \epsilon^{3/2} N} \text{ as required.}$$

$$\lim_{N \rightarrow \infty} \left(\frac{1}{N} + \frac{1}{N^2} + \dots \right) \approx \frac{1}{N}$$